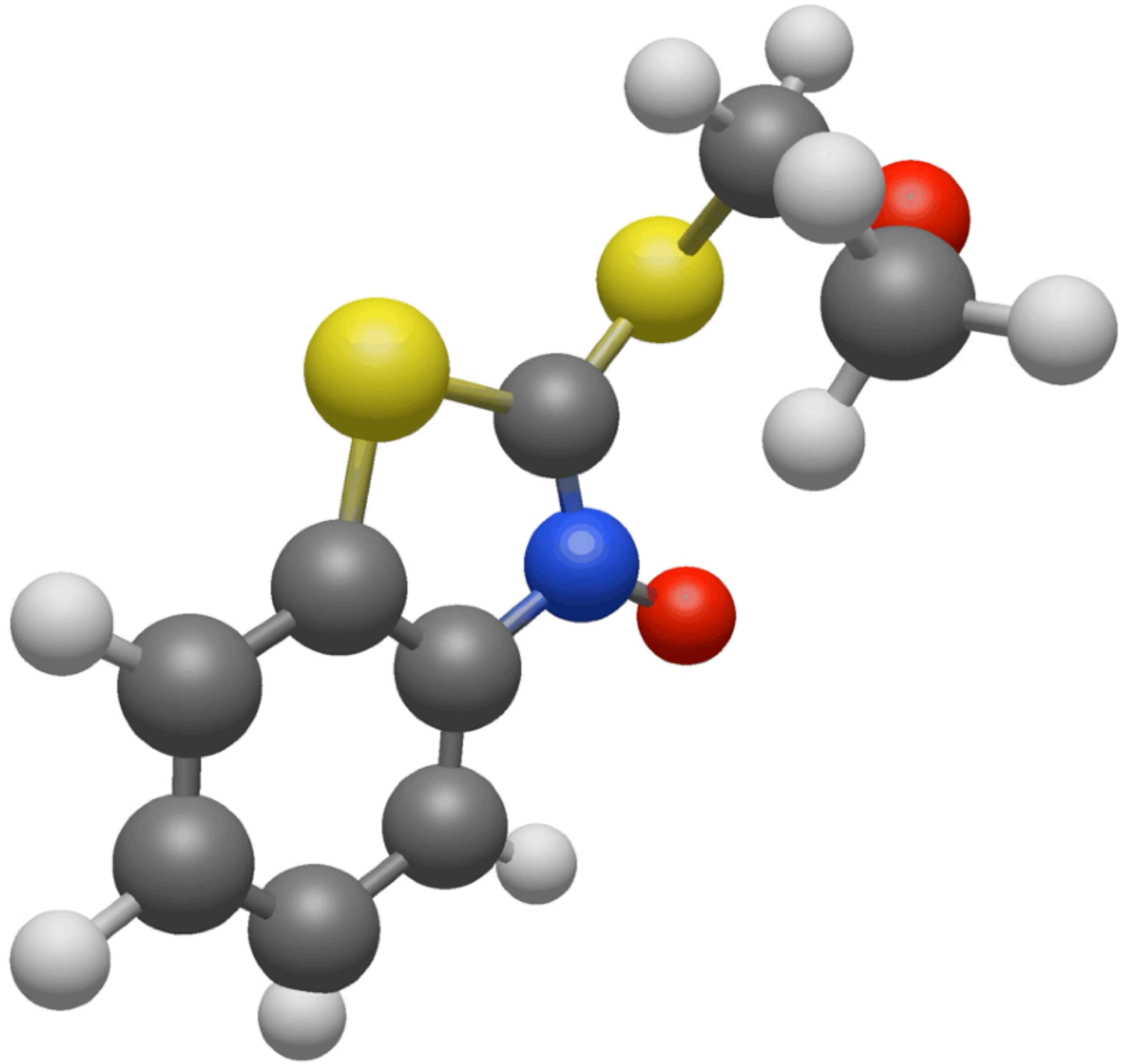


$$\exp(-\tfrac{1}{2}\|x_1 - x_2\|^2) = \exp(x^\top R x), \quad \text{where } R = \begin{bmatrix} 1 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & 0 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 1 \end{bmatrix}$$

31

Here:

-Cartisian gen: $t = 0$ (harmonic prior), e.g., graph GP: 2-particle in 3D, DW-4, LJ-15/55

-torsional generation: (tori space)

-EGNN / tensor field network policy

-**directly from energy without data** (SOC)

Our work:

-more general topology distributions

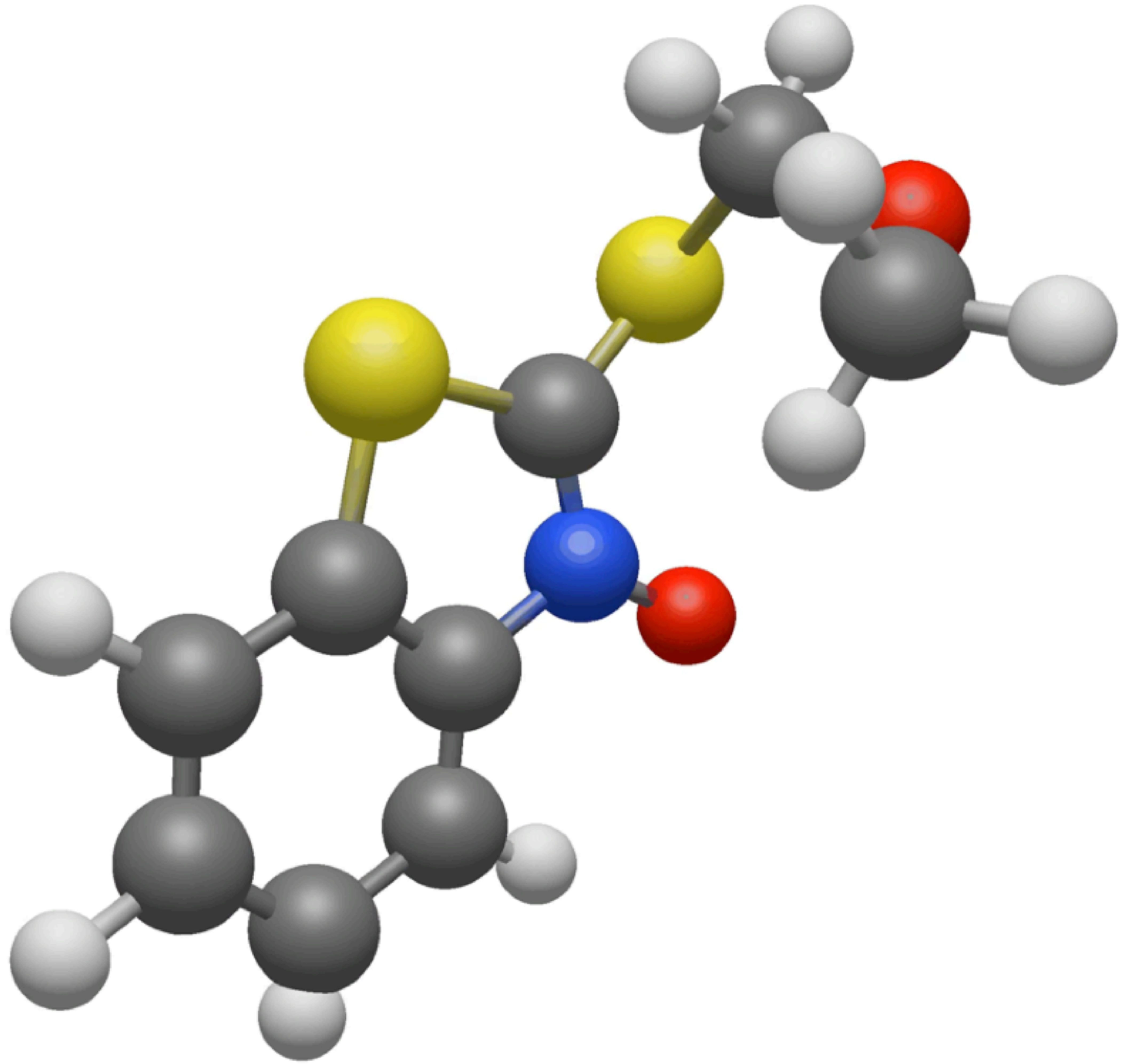
-Topological stochastic heat diffusion

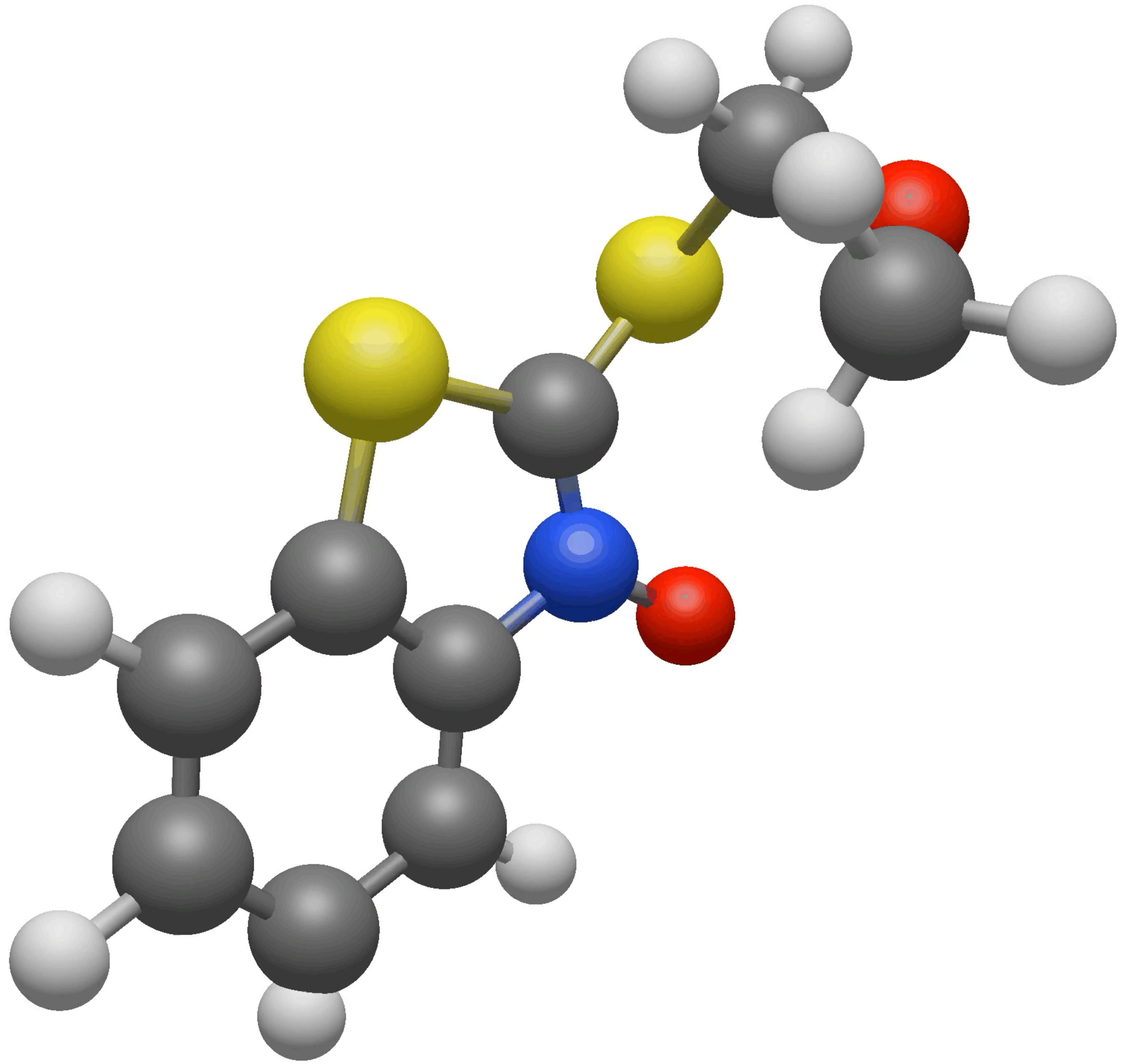
$$dY_t = -cLY_t dt + g_t dW_t$$

-Topology-aware transition kernels

Similar work 2 for conformer generation

[Liu et al. 2025. Adjoint SB Sampler. Neurips] [Haven et al. Adjoint sampling. ICML 2025]





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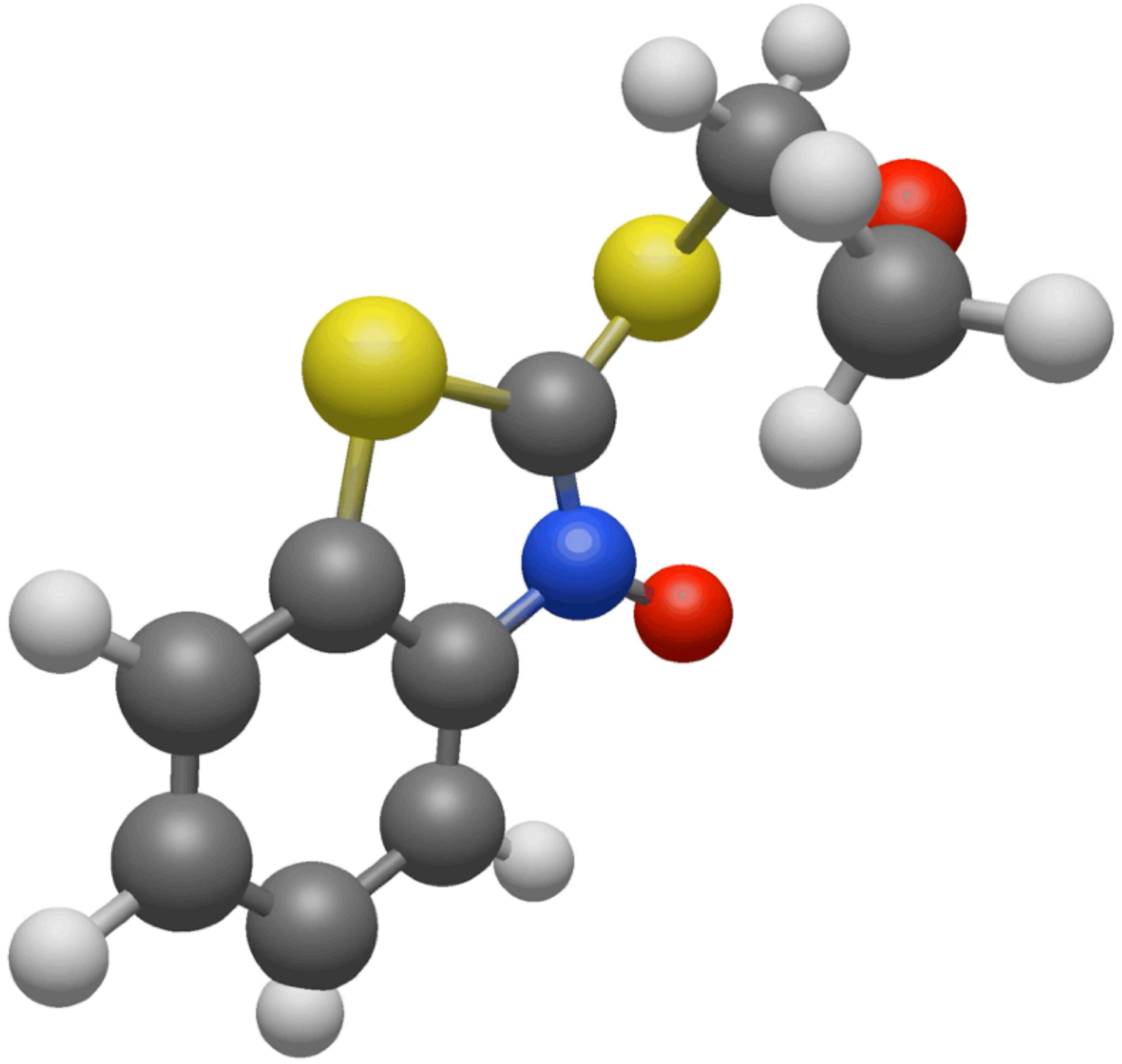

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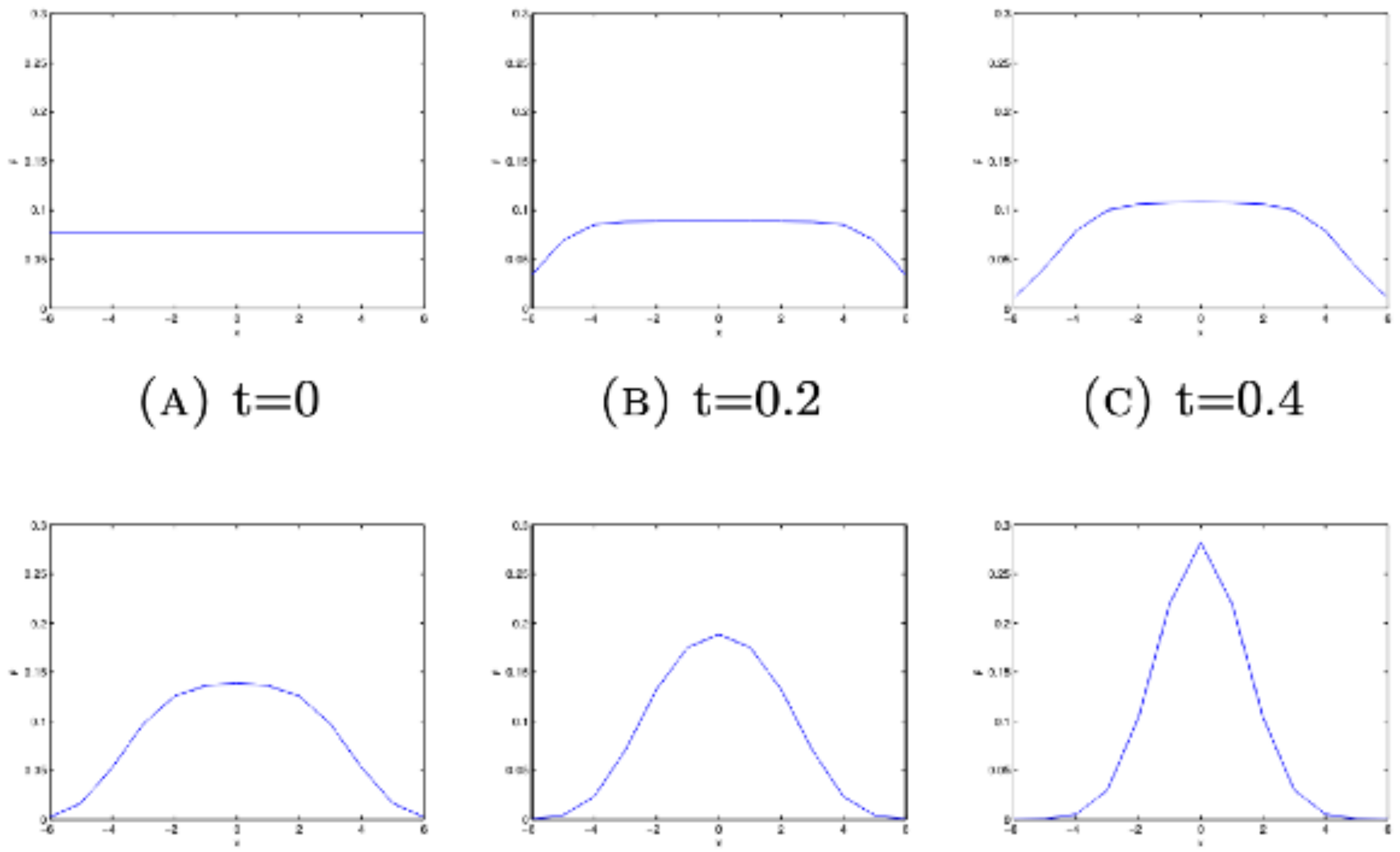
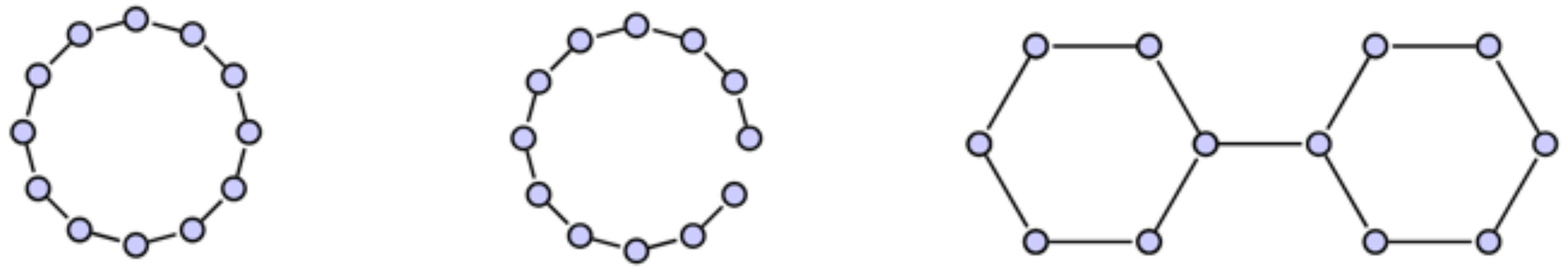
Related work: SBP or OT on graphs for discrete state space



(A) t=0 (B) t=0.2 (C) t=0.4

(D) t=0.6 (E) t=0.8 (F) t=1



29.2 30.5 8.0

- Graphs as samples of discrete distributions on node state space $\sum p_i = 1$
- CTMC as the stochastic process
- SBP / OT on graphs, SOC form
  $$\min_{p,b} \int_0^1 \frac{1}{2} \|b\|^2 dt \quad \text{s.t.} \quad b - \text{controled Kolmogorov}$$
- Connected to Boltzmann distribution — adjoint sampling?
- Some works on this: graph generation Xu et al. 2025
- CTMC ~~ Hodge decomposition
- For both molecular generation, and protein optimization